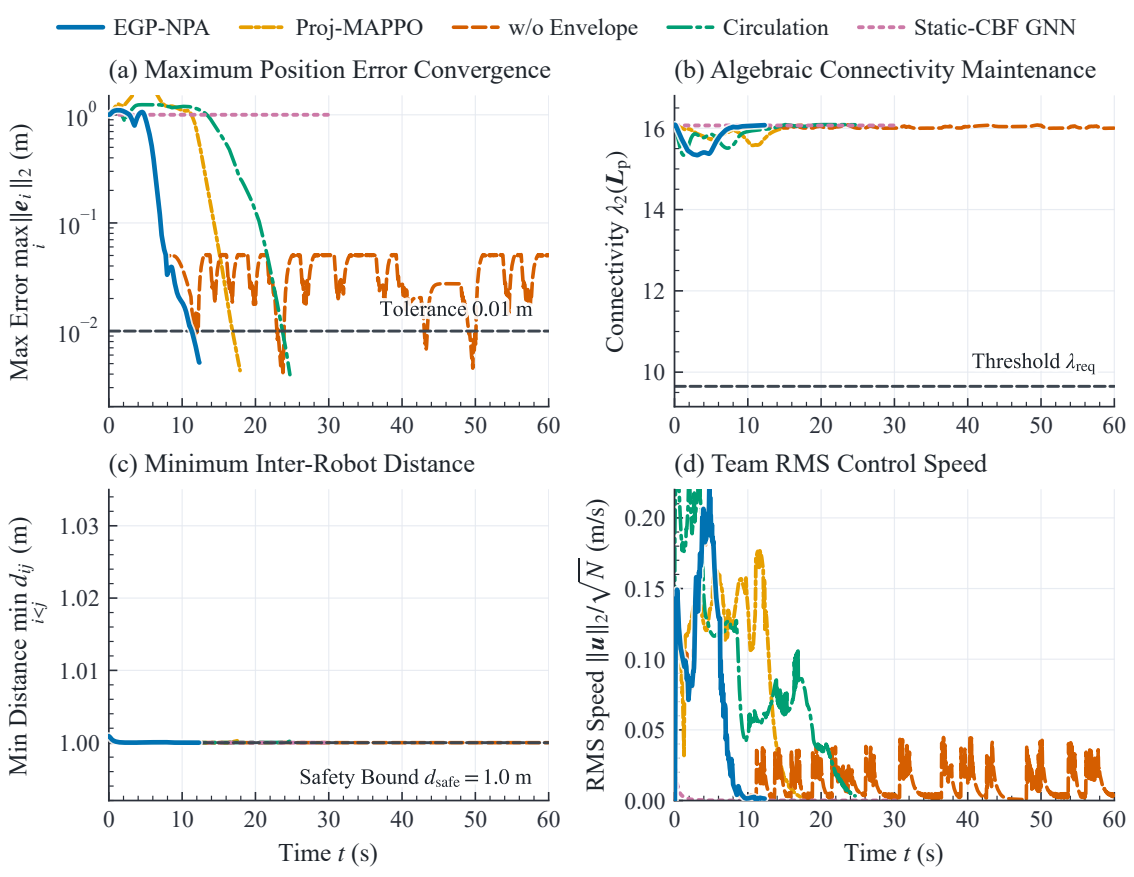

EGP-NPA
Proj-MAPPO
w/o Envelope
Circulation
Static-CBF GNN
(a) Maximum Position Error Convergence
Max Error $\max_i \|e_i\|_2$ (m)
Tolerance 0.01 m
(b) Algebraic Connectivity Maintenance
Connectivity $\lambda_2(L_p)$
Threshold $\lambda_{req}$
(c) Minimum Inter-Robot Distance
Min Distance $\min_{i<j} d_{ij}$ (m)
Safety Bound $d_{safe} = 1.0$ m
Time $t$ (s)
(d) Team RMS Control Speed
RMS Speed $\|u\|_2/\sqrt{N}$ (m/s)
Time $t$ (s)